Course Project Proposal

Project Type: Software Development Project

Name of Project: RuleGuard: Business Rule Conflict Detection and Resolution System

Short Description of Why an Organization Would Find This System Useful

I will develop a Python-based application that helps organizations check their business rules before using them in real processes. Organizations often maintain rules from different departments, and some of these rules may overlap, contradict each other, or contain conditions that can never be reached. I will build a system that detects these problems and explains the exact conditions causing them, so that users can review and correct the rules before they create incorrect decisions.

Description of Organizational Requirements of the System

I will design the system to allow users to enter, edit, and organize rules using a structured format such as conditions, operators, values, and expected decisions. The system will use a rule-processing algorithm to compare rules, identify overlapping conditions, detect contradictory decisions, and find unreachable rules. It will also generate test cases for important combinations of conditions and show which rules would be applied. I will provide a Streamlit interface where users can review conflicts, inspect the affected rules, and mark issues as resolved. The system will use local data storage and will not require external services.

Expected Advantages That the System Will Provide

I expect the system to help organizations find rule problems earlier, reduce incorrect decisions caused by conflicting policies, and make rule maintenance easier. It will provide clear explanations instead of only showing that a conflict exists. It will also help users test changes to rules and understand whether a new rule affects existing decisions. This can reduce manual checking and improve confidence in the consistency of organizational policies.

Any Special Requirements or Resources Anticipated

I will use Python, Streamlit, SQLite, and free Python libraries for the implementation. I will develop and demonstrate the application on my laptop using locally created sample rule sets. I will implement the rule comparison and conflict-detection logic myself and test it with different rule combinations, including overlapping, contradictory, and unreachable rules.

LLM Used: ChatGPT

Prompt 1:

I need to develop a semester-long software development project using Python or Java. I want the project to solve a real problem that an organization may face. Please suggest some practical project ideas that are realistic to develop and demonstrate during one semester.

ChatGPT Response:

One possible project is a Business Rule Conflict Detection System. The system could help an organization identify business rules that overlap, contradict each other, or contain conditions

that cannot be reached. A Python application could allow users to enter structured rules and check them for conflicts before the rules are used in business processes.

Prompt 2:

I am interested in the Business Rule Conflict Detection System. I want to make it a Python-based project. Help me develop the idea further and explain what the system should do.

ChatGPT Response:

The system could allow users to enter and organize business rules using conditions, operators, values, and expected decisions. The application could compare rules and identify overlapping conditions, contradictory decisions, and unreachable rules. It could also generate test cases for important combinations of conditions and show which rules would be applied.

Prompt 3:

I want to use Python, Streamlit, and SQLite for this project. Please make the project realistic for a semester-long software development project and explain what the user interface and system should include.

ChatGPT Response:

A Streamlit interface could allow users to enter, edit, and review business rules. The application could display detected conflicts and show the rules and conditions involved. Users could also review generated test cases and mark identified issues as resolved. SQLite could be used for local storage so the project does not require external services.

Prompt 4:

Please help me identify the main advantages and special requirements for this business rule conflict detection project so I can include them in a course project proposal.

ChatGPT Response:

The system could help organizations find rule problems earlier, reduce incorrect decisions

caused by conflicting policies, and make rule maintenance easier. A useful feature would be to explain why a conflict exists instead of only reporting that one exists. The project could be developed using Python, Streamlit, SQLite, and free Python libraries on a local computer with sample rule sets for testing.